Practical exercise 1

1. Q-Q plots can be used, as shown in lectures, to investigate how two distributions (e.g. data, theoretical) differ in location, spread or shape. In the following examples you can explore the fit of the normal distribution to data simulated from various distributions.

   We start with a sample generated from $N(0,1)$ and we want to verify that $N(0,1)$ is a suitable distribution for these data. First simulate the sample (of size, say, 200) and then produce the normal Q-Q plot. The following `R` commands will do this:

   ```
   > x1 = rnorm(200)
   > qqnorm(x1, main="QQ plot of N(0,1) sample of size 200", pch=19, col="red")
   ```

   Check the produced plot and make sure you understand how the `main`, `pch` and `col` arguments are used. In `R`, type

   ```
   > ?qqnorm
   > ?par
   ```

   for help. The latter will be particularly helpful for setting graphical arguments in `R` throughout this course.

   Does it look like the data come from the $N(0,1)$ distribution? Add the $y = x$ line on your plot and comment (again check how the command below works).

   ```
   > abline(0, 1, lty=2, col="blue")
   ```

   You can also produce a histogram of the data and superimpose the $N(0,1)$ density for checking. In `R` start with the histogram

   ```
   > hist(x1, freq=F, xlim=c(min(x1)-1,max(x1)+1),col="red",
   main="N(0,1) density & N(0,1) sample")
   ```

   then determine a grid of suitable values on which the $N(0,1)$ density will be evaluated

   ```
   > grid = seq(-4,4,length=100)
   ```

   and finally draw the pdf line

   ```
   > lines(grid, dnorm(grid), col = "blue")
   ```

   Note that if you wanted the two graphs that you have drawn so far to be on the same `R` window, you could use the following `R` command at the beginning:

   ```
   > par(mfrow=c(1,2))
   ```

   Question: Now you should repeat the procedure above to try the following scenarios:

   - Generate data from $N(-2,1)$ and check how well $N(0,1)$ fits these data;
   - Generate data from $N(0,3)$ and check how well $N(0,1)$ fits;
   - Generate data from $N(-2,3)$ and check how well $N(0,1)$ fits;
   - Generate data from $t(3)$ and check how well $N(0,1)$ fits;
   - Generate data from $Gamma(2,1)$ and check how well $N(2,2)$ fits.

   In each case comment on the fit of the theoretical distribution to the data

2. The dataset `ozone` considered in the lectures may be found on the course data sets web page on Canvas. It gives maximum daily ozone concentrations, in parts per billion, measured at Stamford, Connecticut and Yonkers, New York, during the period May – September one year. Data are given only for the 132 days (out of a total of 152) on which readings were successfully obtained at both locations.

   The Stamford data are being considered in the lectures so we will first consider the set `yonkers`. Download the ozone data, file `ozone.R`, and load into `R` using the `dget` command — see 'Notes on using `R`' on Canvas. After loading, remember to `attach` the data. Then you can see it by just typing

   ```
   > yonkers
   ```

   Use simple numerical and graphical summaries to investigate the distribution of the variable `yonkers`. Examples you might wish to try are `summary, sd, quantile, hist, boxplot` and `stem`, e.g.

   ```
   > summary(yonkers)
   ```

   and so on.

   Before you start producing plots as suggested below, you may find it useful to be able to plot multiple graphs on the same window. For example, you can plot 4 graphs in a 2 by 2 layout by first using the `R` command

   ```
   > par(mfrow=c(2,2))
   ```

   Generate an idealised sample of the same size as `yonkers` from the normal distribution. Do this by using probability points. In `R` this involves the following code:

   ```
   > ppoints(132)
   > qnorm(ppoints(132))
   ```

   Now do a plot of this idealised sample against the actual (sorted) sample. Use the code:

   ```
   > plot(qnorm(ppoints(132)),sort(yonkers))
   ```

   Note that you can also use the following code to obtain the same plot:

   ```
   > plot(sort(yonkers)~qnorm(ppoints(132)))
   ```

   This is actually a Q-Q plot which can more easily be obtained using

   ```
   > qqplot(qnorm(ppoints(132)),yonkers)
   ```

   or simply

   ```
   > qqnorm(yonkers)
   ```

   Try all these different commands and use the help menu in `R` to check how they work. Then make sure that you add a title and axes labels to your plot.

   How close is the graph to a straight line?

   You can see the 'best' line by using the command:

   ```
   > qqline(yonkers)
   ```

   Use Q-Q plots to find a transformation of this variable such that its observations might be more reasonably modelled by a normal distribution (for example try squares, square roots, logs etc).

   Finally, compare the data for Stamford and Yonkers by using a boxplot.

3. In 1798 Cavendish made 29 determinations of the density of the earth (relative to that of water), and obtained the following values:

5.50 5.61 4.88 5.07 5.26 5.55 5.36 5.29
5.58 5.65 5.57 5.53 5.62 5.29 5.44 5.34
5.79 5.10 5.27 5.39 5.42 5.47 5.63 5.34
5.46 5.30 5.75 5.68 5.85

The data are available on Canvas (`reldens.R`) and you should first read them into `R` using the `dget` command (call the data set `rel.dens`).

View the data using:

```
> rel.dens
```

Use the `R` function `summary` to provide a concise description of the data. Use elementary graphical techniques to investigate more carefully the distribution of the data.

Construct an appropriate Q-Q plot to examine whether the data may reasonably be modelled as independent identically distributed observations from a normal distribution, and if so use the plot to determine the mean and standard deviation of an appropriate normal distribution. You can estimate these two parameters by using the equation of the line that is obtained by the `qqline` function. This is best done manually in the following way.

The line drawn by the computer is $y = a + bx$ which is such that the theoretical lower and upper quartiles match those of the data. Write down these values for your sample data (use the summary results). These are the y co-ordinates.

$Q1 = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$ $Q3 = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$

The corresponding x co-ordinates (theoretical quartiles from the idealised sample) are obtained by:

```
> qnorm(0.25)
```

and

```
> qnorm(0.75)
```

Calculate the gradient, b, of the line using the standard equation $y_2 - y_1 = b(x_2 - x_1)$. You should now be able to calculate the intercept, a, using the equation of the line.

It will be shown in lectures that a is the mean of the corresponding normal distribution and b is the standard deviation.

Compare these with the sample mean and standard deviation.

Further, after his first six determinations, $(5.50, 5.61, 4.88, 5.07, 5.26, 5.55)$, Cavendish made improvements to his apparatus. Is there evidence in the data to suggest that his improvements were worthwhile? (Hint: repeat the analysis with these 6 observations omitted.)